

An Indian-born physicist Narinder Kapany is regarded as the *Father of Fibre Optics*.

Points to Remember

- Light is a form of energy which produces the sensation of sight.
- Laws of reflection: i) Angle of incidence is equal to the angle of reflection ii) The incident ray, the normal to the point of incidence and the reflected ray, all lie in the same plane.
- The distance between the pole and the principal focus of the spherical mirror is called focal length. $f = \frac{R}{2}$; where R is the radius of curvature of the mirror.
- Mirror equation: The relation between u, v and f of a spherical mirror is known as mirror formula $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$
- Magnification: m = $\frac{\text{height of the image } h_2}{\text{height of the object } h_1}$
- Laws of refraction: The incident ray, the refracted ray and the normal to the surface separating two medium lie in the same plane. The ratio of the sine of the incident angle ($\angle$i) to the sine of the refracted angle ($\angle$r) is constant i.e. $\mu = \frac{\sin i}{\sin r}$ = constant
- The bending of light when it passes obliquely from transparent medium to another is called refraction.
- When the angle of incidence exceeds the value of critical angle the refracted ray is impossible. Since r > 90° refraction is impossible and the ray is totally reflected back to the same medium (denser medium). This is called as total internal reflection.

A-Z GLOSSARY

Spherical Mirror	A reflecting surface which is a part of a sphere whose inner or outer surface is reflecting.
Concave Mirror	Part of a hollow sphere whose outer part is silvered and/or inner part is the reflecting surface.
Convex Mirror	Part of a hollow sphere whose inner part is silvered and/or outer part is the reflecting surface.
Centre of curvature	The centre of the hollow sphere of which the spherical mirror forms a part.
Radius of curvature	The radius of the hollow sphere of which the spherical mirror forms a part.
Pole	The midpoint of the spherical mirror.
Aperture	The diameter of the circular rim of the mirror.
Principal axis	The normal to the centre of the mirror is called the principal axis.
Principal focus	The point on the principal axis of the spherical mirror where the rays of light parallel to the principal axis meet or appear to meet after reflection from the spherical mirror.

IX_SCI_EM_Unit-06_PHY.indd 76 12-02-2020 14:16:47